EXPLAINERS (HTTPS://ELITEDATASCIENCE.COM/CATEGORY/EXPLAINERS)

WTF is the Bias-Variance Tradeoff? (Infographic) (https://elitedatascience.com/bias-variance-tradeoff)

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Overheard after class: “doesn’t the Bias-Variance Tradeoff sound like the name of a treaty from a history documentary?”

Ok, that’s fair… but it’s *also* one of the most important concepts to understand for supervised machine learning and predictive modeling.

Unfortunately, because it’s often taught through dense math formulas (https://en.wikipedia.org/wiki/Bias%E2%80%93variance_tradeoff), it’s earned a tough reputation. But as you’ll see in this guide, it’s not that bad. In fact, the Bias-Variance Tradeoff has simple, *practical* implications around model complexity, over-fitting, and under-fitting.

BIAS

VERSUS

VARIANCE

ELITEDATASCIENCE

WTF is the Bias-Variance Tradeoff?

Need to train a predictive model?

If so, you have many supervised learning algorithms to choose from! i.e. Regressions, Decision Trees, Neural Networks, SVM's, etc...

Well, those algos are distinct...

...in many ways. But a major difference is their amount of Bias vs. Variance, which are two types of prediction error.

Bias occurs when an algo has *limited flexibility* to learn the true signal from a dataset.

Variance refers to an algo's *sensitivity* to specific sets of training data.

Wait, "sensitivity" to training data?

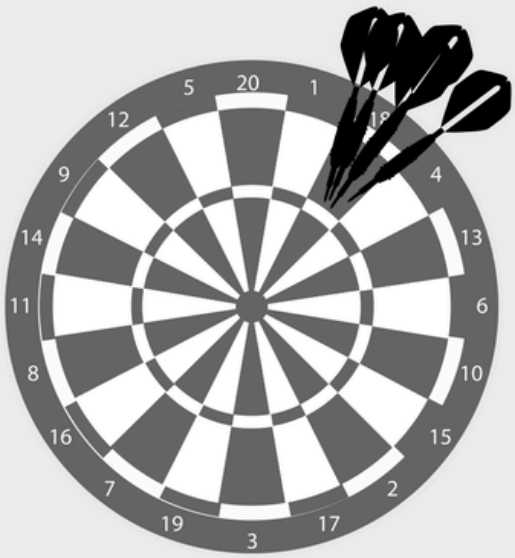
Yes. It's much easier to wrap your head around this concept if you think of algorithms not as one-time methods for training individual models, but instead as repeatable processes.

Ok, let's do a thought experiment...

Imagine you've collected 5 different training datasets for the same problem. Now, imagine using one algorithm to train 5 models (one for each training set).

Here's what those 5 models tell you about your chosen algorithm:

High Bias
Low Variance



High bias, low variance algorithms train models that are consistent, but inaccurate *on average*.

High Variance
Low Bias

High variance, low bias algorithms train models that are accurate *on average*, but inconsistent.

But why is there a tradeoff?

Low variance algos tend to be **less complex**, with simple or rigid underlying structure.

- e.g. Regression
- e.g. Naive Bayes
- *Linear algos*

Low bias algos tend to be **more complex**, with flexible underlying structure.

- e.g. Decision trees
- e.g. Nearest neighbors
- *Non-linear algos*

- *Parametric algos*
- *Non-parametric algos*

Within each algo family, there's a tradeoff too...

For example, regression can be **regularized** to further reduce complexity.

For example, decision trees can be **pruned** to reduce complexity.

That's why a solid model training methodology is key.

Algos that are not complex enough produce **underfit** models that can't learn the signal from the data.

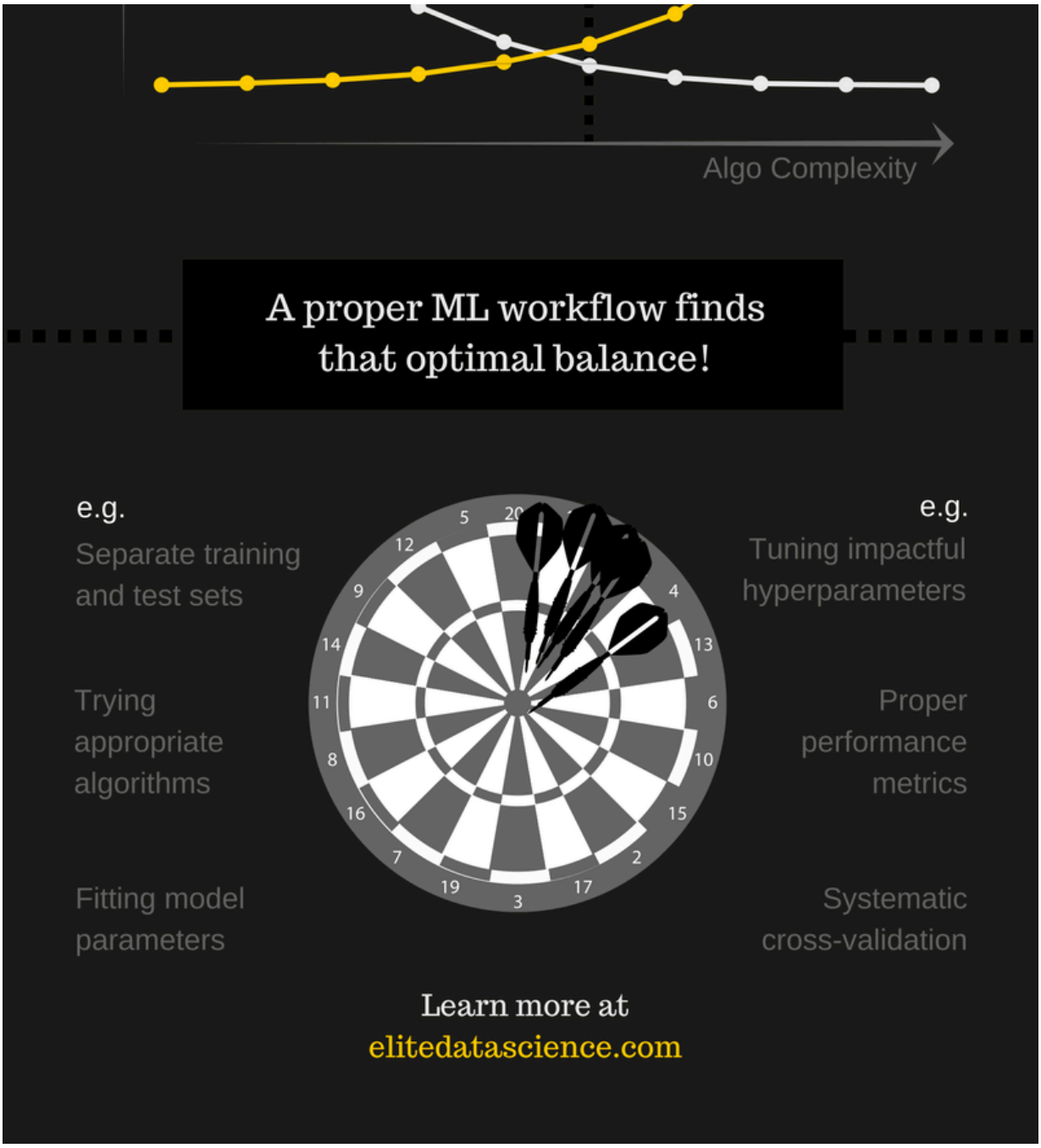
Algo that are too complex produce **overfit** models that memorize the noise instead of the signal.

To get good predictions, you'll need to find a balance of Bias and Variance that minimizes total error.

Total error breaks down as:

Total Error = Bias^2 + Variance + Irreducible Error

(Irreducible error is "noise" that can't be reduced by algorithms. It can sometimes be reduced by better data cleaning)



Share this Infographic on the Bias-Variance Tradeoff:

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Here's our take on the insights from the infographic.

Supervised Learning

The Bias-Variance Tradeoff is relevant for supervised machine learning – specifically for predictive modeling. It's a way to diagnose the performance of an algorithm by breaking down its prediction error.

In machine learning, an **algorithm** is simply a repeatable process used to train a **model** from a given set of **training data**.

- You have many algorithms to choose from, such as Linear Regression, Decision Trees, Neural Networks, SVM's, and so on.
- You can learn more about them in our practical tour through modern machine learning algorithms (https://elitedatascience.com/machine-learning-algorithms).

As you might imagine, each of those algorithms behave very differently, each shining in different situations. One of the key distinctions is how much bias and variance they produce.

There are 3 types of prediction error: bias, variance, and irreducible error.

Irreducible error is also known as "noise," and it can't be reduced by your choice in algorithm. It typically comes from inherent randomness, a mis-framed problem, or an incomplete feature set.

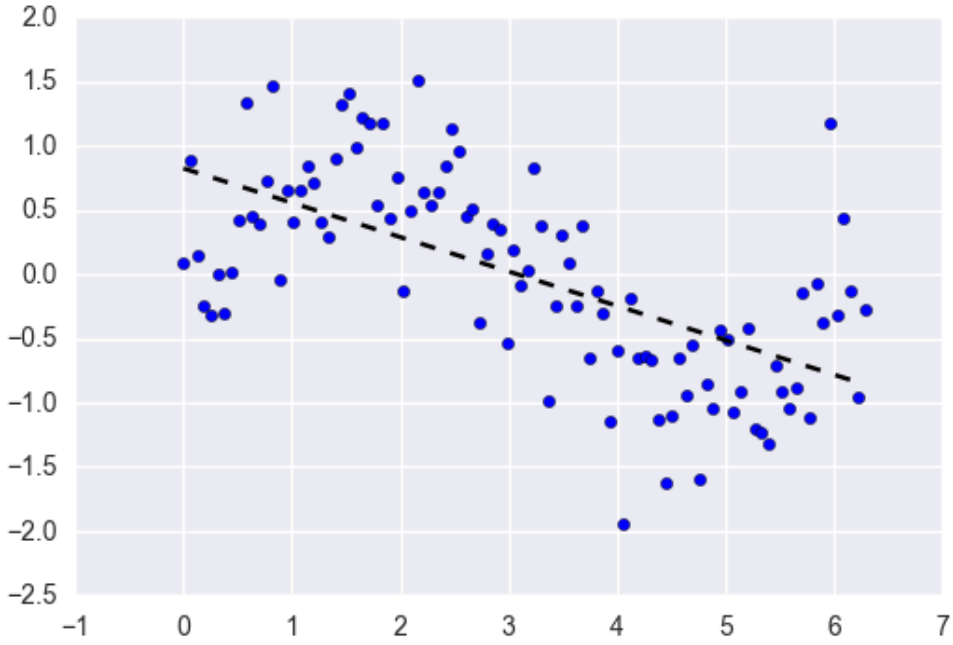
The other two types of errors, however, can be reduced because they stem from your algorithm choice.

Error from Bias

Bias is the difference between your model's expected predictions and the true values.

That might sound strange because shouldn't you "expect" your predictions to be close to the true values? Well, it's not always that easy because some algorithms are simply too rigid to learn complex signals from the dataset.

Imagine fitting a linear regression to a dataset that has a non-linear pattern:



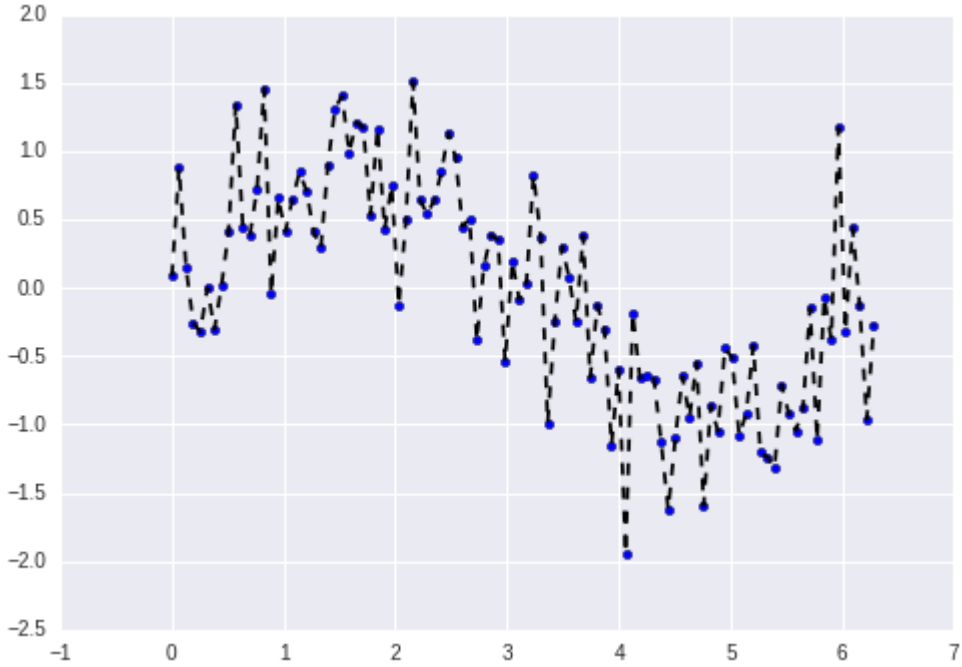
No matter how many more observations you collect, a linear regression won't be able to model the curves in that data! This is known as **under-fitting**.

Error from Variance

Variance refers to your algorithm's sensitivity to specific sets of training data.

High variance algorithms will produce drastically different models depending on the training set.

For example, imagine an algorithm that fits a completely unconstrained, super-flexible model to the same dataset from above:



As you can see, this unconstrained model has basically memorized the training set, including all of the noise. This is known as **over-fitting**.

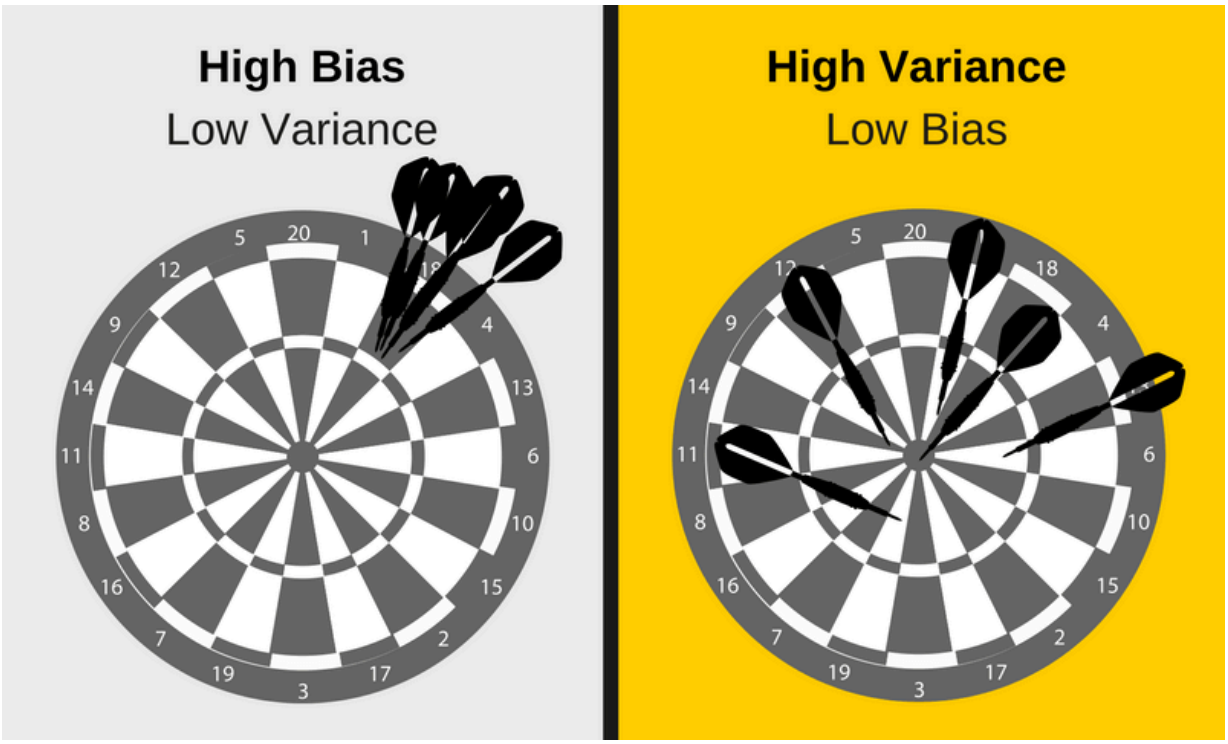
The Bias-Variance Tradeoff

It's much easier to wrap your head around these concept if you think of algorithms not as one-time methods for training individual models, but instead as repeatable processes.

Let's do a thought experiment:

1. Imagine you've collected 5 *different* training sets for the *same* problem.
2. Now imagine using one algorithm to train 5 models, one for each of your training sets.
3. Bias vs. variance refers to the accuracy vs. consistency of the models trained by your algorithm.

We can diagnose them as follows.



Low variance (high bias) algorithms tend to be **less complex**, with simple or rigid underlying structure.

- They train models that are consistent, but inaccurate *on average*.
- These include linear or parametric algorithms such as regression and naive Bayes.

On the other hand, low bias (high variance) algorithms tend to be **more complex**, with flexible underlying structure.

- They train models that are accurate *on average*, but inconsistent.
- These include non-linear or non-parametric algorithms such as decision trees and nearest neighbors.

This **tradeoff in complexity** is why there's a tradeoff in bias and variance – an algorithm cannot simultaneously be more complex and less complex.

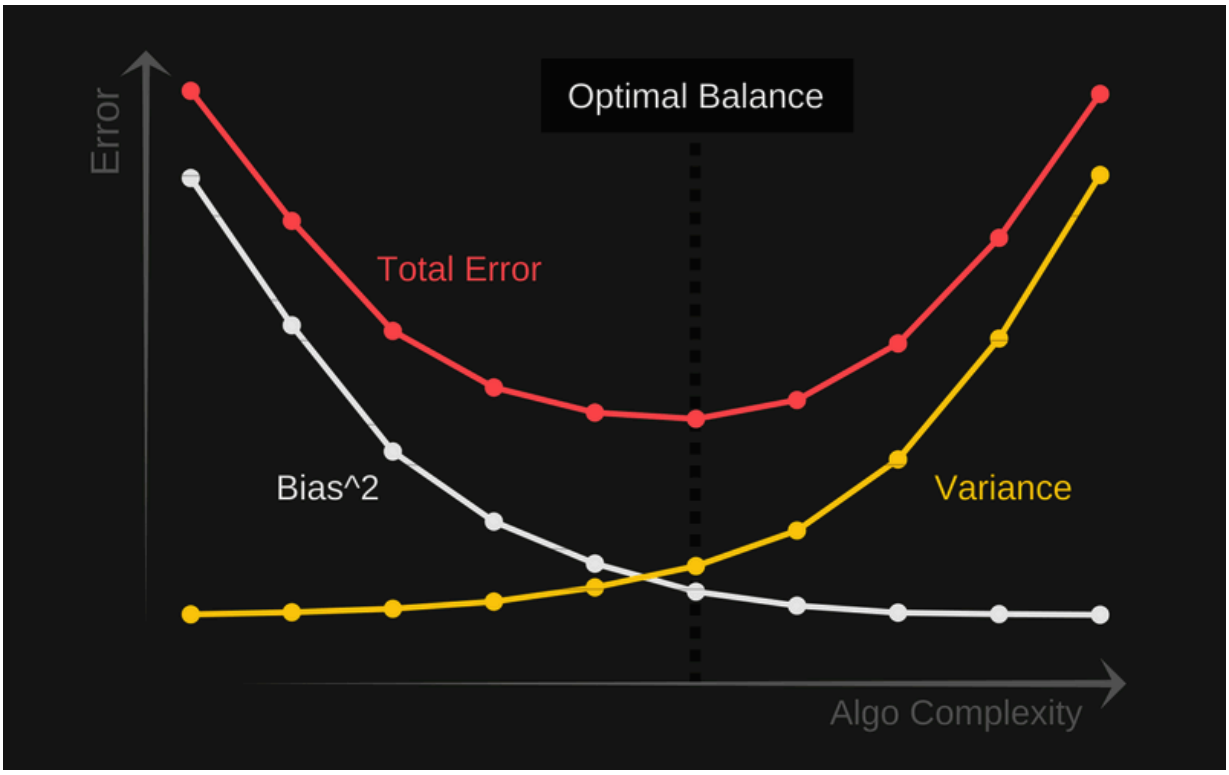
**Note: For certain problems, it's possible for some algorithms to have less of both errors than others. For example, ensemble methods (i.e. Random Forests) often perform better than other algorithms in practice. Our recommendation is to always try multiple reasonable algorithms for each problem.*

Total Error

To build a good predictive model, you'll need to find a balance between bias and variance that minimizes the total error.

Total Error = Bias^2 + Variance + Irreducible Error

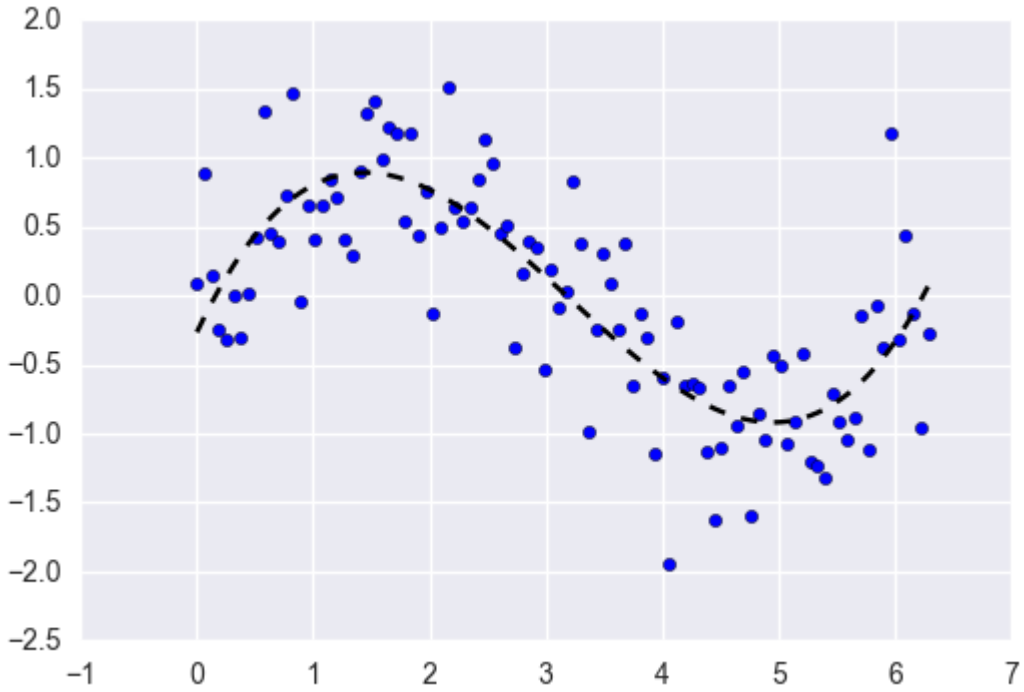
Machine learning processes find that optimal balance:



A proper machine learning workflow includes:

- Separate training and test sets
- Trying appropriate algorithms (No Free Lunch (https://elitedatascience.com/machine-learning-algorithms#no-free-lunch))
- Fitting model parameters
- Tuning impactful hyperparameters
- Proper performance metrics
- Systematic cross-validation

Finally, as you might have already concluded, an optimal balance of bias and variance leads to a model that is neither overfit nor underfit:



This is the ultimate goal of supervised machine learning – to isolate the **signal** from the dataset while ignoring the noise!

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